Characterizing the Optimal $\mathbf{0-1}$ Loss for Multi-class Classification with a Test-time Attacker

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Motivation

- ML models are vulnerable to **adversarial examples** -- can be harmful in safety critical applications
- The **optimal loss** is a diagnostic tool to
  - determine the robustness of the best possible classifier
  - compare with state-of-the-art robust training methods (e.g. adversarial training)

Can we provide a tight lower bound on loss in the presence of a test-time attacker?

Notation	Explanation
h	soft classifier
P	discrete data distribution with finite support
ℓ	$\mathbf{0-1}$ loss function: $\ell(h,\tilde{x},y) = 1 - h(x)_y$
$N(x)$	adversarial neighborhood (L_2 ball around x)
p	probability mass vector for P
$q(h)$	vector of $\min_{\tilde{x}\in N(x)} h(\tilde{x})_y$
ϵ	perturbation constraint

Formulating Lower Bounds as a Linear Program (LP)

- We are interested in the lower bound of the optimal adversarial loss:

$$L^* = \min_h \mathbb{E}_{(x,y)\sim P}\left[\max_{\tilde{x}\in N(x)} \ell\big(h,(\tilde{x},y)\big)\right]$$

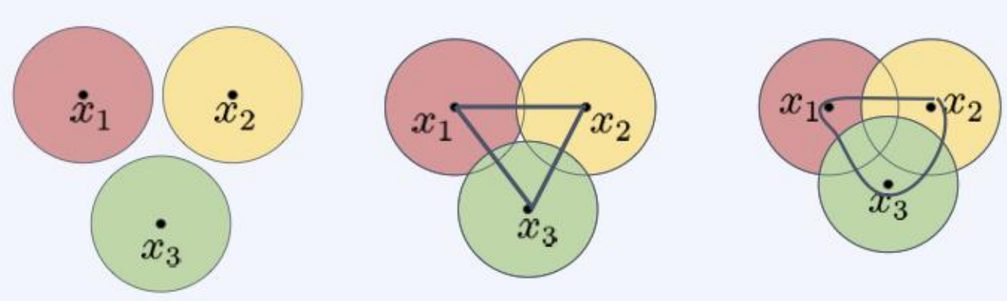
- Since $\ell(h,\tilde{x},y) = 1 - h(x)_y$, we have

$$L^* = 1 - \max_h \mathbb{E}_{(x,y)\sim P}\left[\min_{\tilde{x}\in N(x)} h(\tilde{x})_y\right]$$

$$= 1 - \boxed{\max_h p^T q(h)}$$

Objective is linear over vectors $q(h)$.

Constructing the constraints of the LP: conflict hypergraph



Each input is a vertex. 3 edges. A degree 3 hyperedge.

Theoretical Result

Lemma 1 - The optimal loss can be written as the solution to a LP :

$$1 - L^* = \max_q p^T q \; s.t. \; q \geq 0, B^{(\leq K)} q \leq 1$$

where

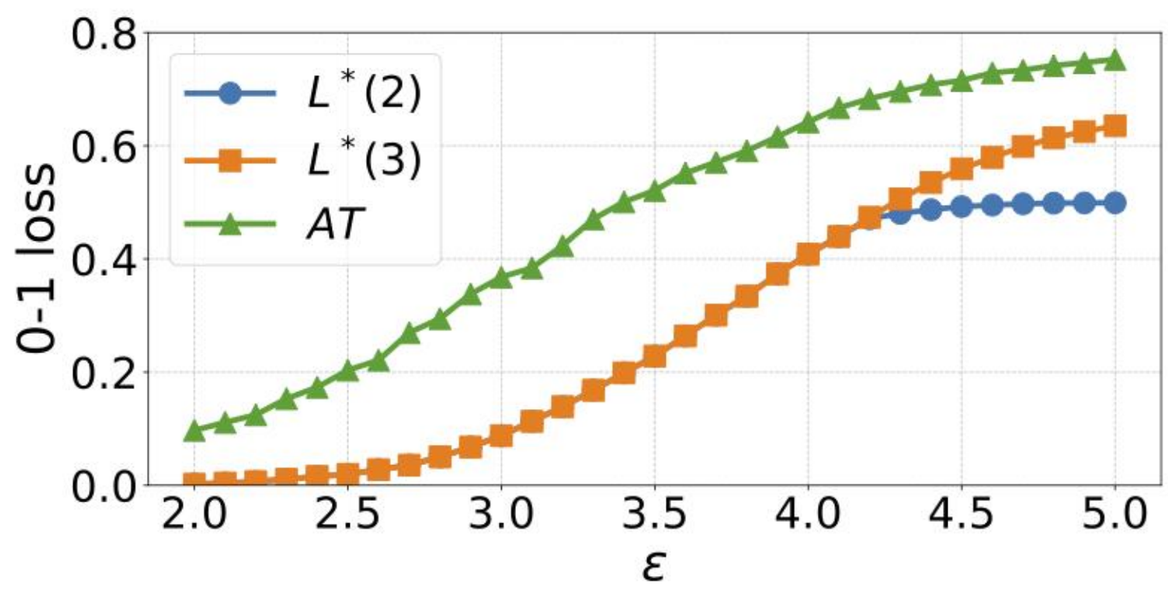
- K is the number of classes
- $B^{(\leq K)}$ is the incidence matrix of the conflict hypergraph considering up to degree K hyperedges

Experimental Results

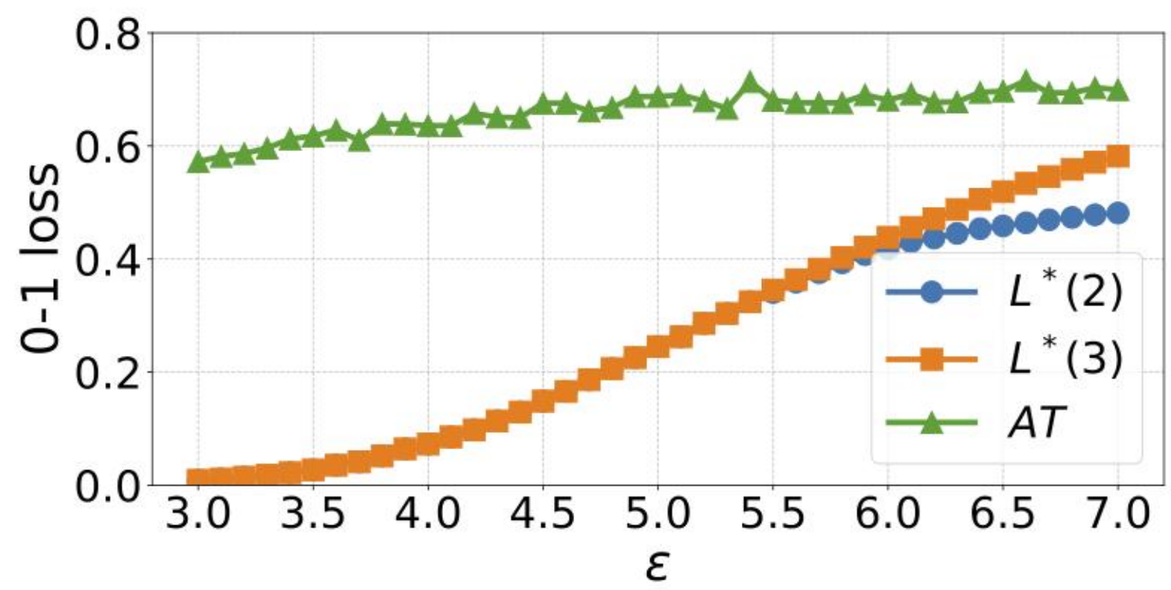
Comparing lower bounds with adversarial training (under L_2 adversary).

- $L^*(2)$: loss of conflict graph contains only edges
- $L^*(3)$: loss of conflict graph contains up to degree 3 hyperedges
- AT: loss of adversarial training

MNIST Class 1, 4, 7



CIFAR-10 Class 0, 2, 8



- **At small ϵ, increasing the hyperedge degree does not affect the lower bound much.**
- **A large gap between the computed lower bound and performance of adversarial training.**

For detailed proofs and more results, see full paper: